

Math 133 Written Homework 5

Derivatives and Instantaneous Rates of Change

Due Thursday, October 15, 2026, at 11:59 p.m.

Show your work, including algebraic simplifications and reasons for your conclusions. List collaborators, time spent, and any use of generative AI with your submission. Your submitted work must be your own. Late work will not be accepted.

Unless otherwise stated, compute derivatives using the limit definition, not differentiation rules. When evaluating limits, justify any use of continuity or limit laws by explaining why the relevant conditions are satisfied. Include units when interpreting rates of change.

1. Use the Intermediate Value Theorem to show that

$$x^3 + x = 10$$

has a solution in $(1, 3)$. State the function and interval to which you apply the theorem, and justify its hypotheses.

2. State the limit definition of $f'(a)$. Explain the geometric meaning of the difference quotient in this definition and of its limit. What must be true of the limit for $f'(a)$ to exist?
3. Let $f(x) = 7 - 3x$. Find $f'(x)$ using the limit definition, and explain how your answer relates to the graph of f .

4. Let $g(x) = x^2 - 4x + 1$. Find $g'(x)$ using the limit definition. Find an equation of the tangent line to the graph of g at $x = 3$.
5. Let $r(x) = \sqrt{x + 5}$. State the domain of r , and compute $r'(4)$ using the limit definition. Write an equation of the tangent line at $x = 4$.
6. A differentiable function F satisfies $F(2) = 5$ and $F'(2) = -3$. Find an equation of the tangent line at $x = 2$, and use it to estimate $F(2.1)$. Does your value necessarily agree with $F(2.1)$? Explain why or why not.

7. A power plant models the cost of operating at an output of q kilowatts for a fixed one-hour period by

$$C(q) = 1200 + 8q + 0.02q^2, \quad 0 \leq q \leq 500,$$

where $C(q)$ is measured in dollars.

- Compute $C'(100)$. Use any method you want.
 - Find the tangent line $L(q)$ to the graph of C at $(100, C(100))$, and evaluate $L(100.5)$.
 - Use your tangent line to estimate the additional cost of increasing output from 100 to 100.5 kilowatts. Compare this with the actual difference $C(100.5) - C(100)$. How is your estimate related to $C'(100)$, and why is it different from $L(100.5)$?
8. A company has differentiable cost and revenue functions $C(q)$ and $R(q)$, measured in dollars, where q is the number of goods produced and sold. Its profit is $P(q) = R(q) - C(q)$. Suppose

$$C(200) = 3400, \quad R(200) = 5000, \quad C'(200) = 12, \quad R'(200) = 18.$$

Complete the table. Use tangent-line estimates where indicated, and include units. The additional amount for the 201st good is the change in the corresponding total from $q = 200$ to $q = 201$; the data allow you to estimate this actual change, not determine it exactly.

Quantity	Cost	Revenue	Profit
Total at $q = 200$			
Marginal value at $q = 200$			
Estimated total at $q = 201$			
Estimated additional amount for the 201st good			